

EAS 6134: Inverse Methods in EAS
Homework 2: Norms, Inner Products, and a First Inverse Problem
due Wednesday February 4

A normed vector space is a linear vector space, V , in which for every elements $f \in V$ a functional called the norm is defined and written as $\|f\|$. If $f, g, h \in V$ and $\alpha \in \mathbb{R}$ then

$$\begin{aligned}\|f\| &\geq 0 \\ \|\alpha f\| &= |\alpha| \|f\| \\ \|f + g\| &\leq \|f\| + \|g\| \\ \|h\| &= 0 \text{ only if } h = 0\end{aligned}$$

A useful generalization of the dot product between 2 vectors is the inner product. A real inner product space is a real linear vector space, V , in which for every pair of elements f, g , a functional called the inner product is defined and written as (f, g) . If $f, g, h \in V$ and $\alpha \in \mathbb{R}$ then

$$\begin{aligned}(f, g) &= (g, f) \\ (f + g, h) &= (f, h) + (g, h) \\ (\alpha f, g) &= \alpha(f, g) \\ (f, f) &> 0, \text{ if } f \neq 0\end{aligned}$$

1. For $x \in \mathbb{R}^n$ show that the functional

$$N[x] = [|x_1|^{\frac{1}{2}} + |x_2|^{\frac{1}{2}} + \dots + |x_n|^{\frac{1}{2}}]^2$$

is not a norm for the space when $n > 1$. Hint: treat the case $n = 2$ first.

2. Consider the linear vector space of symmetric matrices $S^{n \times n}$. The *trace* of the matrix $A \in S^{n \times n}$ is given by

$$tr(A) = \sum_{j=1}^n A_{jj}$$

The Frobenius norm of an $m \times n$ matrix is given by

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2}$$

Show that $tr(AB) = (A, B)$ is an inner product on $S^{n \times n}$. Show the norm associated with the inner product is the Frobenius norm.

Exercise 3 of Chapter 1 in ABT

3. Consider the borehole vertical seismic profile problem of Examples 1.3 and 1.9 in ABT for $n = 100$ equally spaced seismic sensors located at depths of $z = 0.2, 0.4, \dots, 20$ m, and for a model $\mathbf{m}$ describing n corresponding equal length seismic slowness values for 0.2 m intervals having midpoints at $z = 0.1$ m.

a. Calculate the appropriate system matrix $\mathbf{G}$ for discretizing the integral equation (1.21) using the midpoint rule.

b. For a seismic velocity model having a linear depth gradient specified by $v = v_0 + kz$ (1.61) where the velocity at $z = 0$ is $v_0 = 1$ km/s and the gradient is $k = 40$ m/s per m, calculate the true slowness values at the midpoints of the n intervals, $\mathbf{m}_{\text{true}}$. Additionally, integrate the corresponding slowness function for (1.61) using (1.21) to calculate a noiseless synthetic data vector, $\mathbf{d}$, of predicted seismic travel times at the sensor depths.

c. Solve for the slowness, $\mathbf{m}$, as a function of depth using your $\mathbf{G}$ matrix and analytically calculated noiseless travel times using the MATLAB backslash or an equivalent operator. Compare your result graphically with $\mathbf{m}_{\text{true}}$.

d. Generate a noisy travel time vector where independent normally distributed noise with a standard deviation of 0.05 ms is added to the elements of $\mathbf{d}$. Re-solve the system for $\mathbf{m}$ and again compare your result graphically with $\mathbf{m}_{\text{true}}$. How has the model changed?

e. Repeat the problem, but for just $n = 4$ sensor depths and corresponding equal length slowness intervals. Is the recovery of the true model improved? Explain in terms of the condition numbers of your $\mathbf{G}$ matrices.